COMP233 – Discrete Mathematics

Chapter 4: Number Theory and Methods of Proof

Lecture 4.4: Quotient-Remainder Theorem

Every integer can be represented uniquely by its quotient and remainder, and this representation provides a powerful method for proving mathematical properties.

1. Divisibility and Motivation

Exact Division

If d divides n ($d \mid n$), then there exists an integer k such that $n = dk$ and the remainder is 0.

When division is not exact

For any integer n and positive integer d, we can still write

$$n = dq + r$$

where r is the amount left over (the **remainder**).

2. Quotient-Remainder Theorem

Theorem 4.4.1 (Quotient-Remainder Theorem)

For every integer n and every positive integer d, there exist unique integers q and r such that

$$n = dq + r, \quad 0 \le r < d.$$

q : Quotient
Number of complete groups of size d in n.

r : Remainder
Amount left over, with $0 \le r < d$.

Uniqueness

The integers q and r are unique. If $n = dq + r$ and $n = dq' + r'$ with $0 \le r, r' < d$, then $q = q'$ and $r = r'$.

3. Computing div and mod

The quotient and remainder are defined by

$$n \text{ div } d = q, \qquad n \bmod d = r$$

where $n = dq + r$ and $0 \le r < d$.

Example: 32 div 9 and 32 mod 9

$$32 = 9(3) + 5$$
$$32 \text{ div } 9 = 3, \qquad 32 \bmod 9 = 5.$$

4. Examples of Quotient and Remainder

n	d	$q = n \text{ div } d$	$r = n \bmod d$	Check
54	4	13	2	$54 = 4(13) + 2$
-54	4	-14	2	$-54 = 4(-14) + 2$
54	70	0	54	$54 = 70(0) + 54$
-17	5	-4	3	$-17 = 5(-4) + 3$

5. Mathematical mod vs. Programming %

In mathematics (Epp), the remainder satisfies $0 \le r < d$.

In languages such as C++ and Java, integer division truncates toward zero and % can produce a negative result for a negative dividend.

Expression	Mathematical result	C++/Java result
$-11 \bmod 4$	1	-3 $(-11 \,\%\, 4)$
$-17 \bmod 5$	3	-2 $(-17 \,\%\, 5)$
$6 \bmod 4$	2	2 $(6 \,\%\, 4)$

6. Applications

Days of the Week

If today is day n, then the day of the week after k days is given by

$$(n + k) \bmod 7.$$

Example
If today is Tuesday ($n = 2$) then the day after 25 days is $(2 + 25) \bmod 7 = 27 \bmod 7 = 6$ which is Saturday.

Modular Arithmetic Problem

Find the remainder when 7^{50} is divided by 5.

Solution
$7 \bmod 5 = 2$. Since $2^4 = 16 \equiv 1 \pmod 5$, we have

$$7^{50} \equiv 2^{50} \equiv (2^4)^{12} \cdot 2^2 \equiv 1 \cdot 4 \equiv 4 \pmod 5.$$

So the remainder is 4.

7. Representing Integers by Remainders

Modulo 2: Even and Odd Integers

Every integer is either

$n = 2q$ (even) or $n = 2q + 1$ (odd)

for some integer q.

These two cases are exhaustive and mutually exclusive.

Modulo 4: Four Possible Remainders

Every integer has exactly one remainder when divided by 4.

Remainder 0	Remainder 1	Remainder 2	Remainder 3
$4q$	$4q + 1$	$4q + 2$	$4q + 3$

These four cases are exhaustive and mutually exclusive.

8. Proof by Division into Cases

1. **Define cases** — List all possible cases (for example, using remainders).
2. **Prove each case** — Prove the statement for each case separately.
3. **Conclude** — Since the cases are exhaustive, the statement is true for all integers.

Example: The square of every odd integer is one more than a multiple of 8.

Prove: $\forall n \in \mathbb{Z}, \quad n \text{ odd} \Rightarrow \exists m \in \mathbb{Z} : n^2 = 8m + 1.$

Proof: An odd integer is either $4q + 1$ or $4q + 3$.

Case 1: $n = 4q + 1$

$$\begin{aligned} n^2 &= (4q + 1)^2 \\ &= 16q^2 + 8q + 1 \\ &= 8(2q^2 + q) + 1. \end{aligned}$$

Case 2: $n = 4q + 3$

$$\begin{aligned} n^2 &= (4q + 3)^2 \\ &= 16q^2 + 24q + 9 \\ &= 8(2q^2 + 3q + 1) + 1. \end{aligned}$$

In both cases the expression in parentheses is an integer. Hence $n^2 = 8m + 1$.

9. Absolute Value

Definition

$$|x| = \begin{cases} x, & x \ge 0 \\ -x, & x < 0 \end{cases}$$

Basic Properties

1. $|x| \ge 0$, and $|x| = 0$ iff $x = 0$.
2. $-|x| \le x \le |x|$.
3. $|-x| = |x|$.

Triangle Inequality

$$|x + y| \le |x| + |y|$$

Proof (by cases on $x + y$):

- If $x + y \ge 0$, then $|x + y| = x + y \le |x| + |y|$.
- If $x + y < 0$, then $|x + y| = -(x + y) = (-x) + (-y) \le |x| + |y|$.

Example
Let $x = -3$, $y = 5$.
$|x + y| = |2| = 2$,
$|x| + |y| = 3 + 5 = 8$,
and $2 \le 8$.

Note
Equality does not always hold. For example, $x = 1$, $y = -1$:
$|x + y| = 0$, $\quad |x| + |y| = 2$.

10. Summary of Conceptual Flow

Divisibility and motivation → Quotient-Remainder Theorem ($n = dq + r$) → div and mod (n div d, n mod d) → Representing integers by remainders (mod 2, mod 4) → Proof by division into cases (exhaustive cases) → Absolute value and its properties

Ruwa' AbuHweidi @ 2026